



SMARTIE

Synergistic Management and Advancement of Artificial
Intelligence in European Higher Education

WP2 – Alliance for Artificial Intelligence in Higher Education – Setup SMARTIE

T2.2. Analysis related to adoption AI in educational process

D.2.2.6. Poll of disciplines and subjects that will be updated

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1. Foreword

The integration of artificial intelligence into higher education is rapidly becoming a strategic priority for universities seeking to modernize teaching, learning, and curriculum development. As AI continues to influence professional practice, research, communication, healthcare, logistics, and engineering systems, higher education institutions are increasingly required to adapt their academic offer in order to prepare students for emerging technological, social, and labour market realities. In this context, the SMARTIE consortium undertook a structured analysis aimed at identifying the disciplines and subjects that can be meaningfully updated through the use of AI tools and AI-supported educational approaches. This report reflects a collaborative effort to explore how AI can enhance subject content, teaching methodologies, and the relevance of university programmes in several key domains. The selected fields—Engineering, Inland Navigation and Logistics, Communication Sciences, and Health and Environment—represent areas in which digital transformation is already generating substantial changes in knowledge requirements, professional competences, and operational practices. By identifying priority subjects for improvement, the consortium contributes to a forward-looking educational strategy that supports innovation, interdisciplinarity, and alignment with societal needs.

The present document therefore serves not only as a record of proposed subjects for update, but also as an important foundation for the next stages of the SMARTIE project, where selected disciplines will be further developed and improved using AI-related tools, methods, and pedagogical perspectives.

2. Introduction

The adoption of artificial intelligence in the educational process has become an increasingly important topic in the modernization of higher education. AI is no longer viewed solely as a technological innovation, but as a transformative factor capable of reshaping how knowledge is produced, taught, learned, and applied in academic and professional contexts. In response to these developments, the SMARTIE project included a dedicated activity focused on identifying the disciplines and subjects that can be updated through the integration of AI tools and concepts into tertiary education.

This report presents the results of a poll and internal analysis carried out within the consortium in order to establish which subjects are most suitable for such an updating process. The analysis covered four broad thematic areas: Engineering, Inland Navigation and Logistics, Communication Sciences, and Health and Environment. For each of these areas, project partners contributed proposals based on their expertise, teaching experience, and understanding of student needs. The report outlines the rationale for integrating AI into these

areas and emphasizes the specific ways in which AI can enrich learning outcomes, strengthen practical relevance, and support the acquisition of future-oriented competences.

In addition, the document explains how the proposed subjects were selected for the next phase of development. Although the consortium initially foresaw a minimum number of disciplines to be proposed by each partner, the final contribution exceeded expectations, resulting in a broader pool of subjects across the four SMARTIE themes. Through an internal survey process among partners, a set of disciplines was selected for update within the next work package. This creates a structured basis for curriculum innovation and for the practical integration of AI into the educational offer of the participating institutions. The topics proposed to be updated, in terms of educational materials, using AI are the following:



Figure 1 Topics proposed for updating using AI support

3. Proposals for subjects and topics

3.1 Engineering

The adoption of artificial intelligence (AI) in tertiary engineering education has become a strategic necessity rather than a technological option. Modern engineering practice increasingly relies on data-driven design, intelligent optimization, automation, and predictive modelling, all of which are underpinned by AI methods. Integrating AI into engineering curricula equips students with the analytical tools required to address complex, multidisciplinary problems and to operate effectively in highly digitalized industrial environments. The connection between engineering and AI is intrinsically bidirectional. Engineering provides the physical systems, mathematical models, and domain knowledge on which AI solutions are built, while AI enhances engineering by enabling smarter design processes, adaptive control systems, advanced simulation, and real-time decision support. In tertiary education, this synergy translates into more relevant learning outcomes, fostering systems thinking, innovation, and problem-solving skills aligned with Industry 4.0 and emerging Industry 5.0 paradigms.

The project partners have expertise in the field of tertiary education related to engineering subjects. Based on their experience and interests for their students, the topic engineering has been divided between 4 partners, each partner contributing to the development of database containing subjects that will be updated or integrated in their curricula using AI technology. Their proposal for engineering topic can be found in Figure 2.

Universitatea din Craiova	Universidade de Lisboa	Politecnico di Bari	Univerza V Ljubljani
• Material Strenght • Welding Technology • Joining Processes of Advanced Materials • Microwave Hybrid Heating of Materials	• Weldability of stainless steels • Weldability of the carbon steels • Adhesive Bonding • Resistance Welding	• Optimised parametric reconfiguration and automation systems • Post processing Additive manufacturing in HRC • Predictive maintenance and tool failure planning • Computer-aided design • Plastic Injection Moulding	• Friction stir welding • Ultrasonic Welding • Active particle TIG welding • Wire-arc additive manufacturing

Figure 2 Engineering

3.2 Inland Navigation and Logistics

The adoption of artificial intelligence (AI) in tertiary education focused on inland waterway transport (IWT), ports, and logistics is becoming strategically important in the context of digital transformation and the transition toward sustainable, data-driven transport systems. Inland navigation and port operations are increasingly dependent on real-time data, predictive analytics, automation, and decision-support systems—core domains of AI application. Therefore, integrating AI into academic programs is essential to prepare future engineers, logisticians, and transport managers for complex, technology-intensive operational environments.

In inland waterway transport, AI supports route optimization based on hydrological data, traffic density, fuel efficiency, and weather forecasting. Machine learning algorithms can predict water levels, congestion risks, and maintenance needs of vessels and infrastructure, thereby enhancing safety and operational reliability. In ports, AI-driven systems enable smart berth allocation, automated cargo handling, predictive maintenance of cranes and handling equipment, and improved terminal throughput. Within logistics, AI enhances supply chain visibility, demand forecasting, multimodal coordination, and inventory optimization.

The connection between IWT, ports, logistics, and AI is therefore systemic: transport and logistics generate large volumes of operational data, while AI provides the analytical and optimization tools necessary to transform this data into actionable intelligence. Embedding

AI competencies in tertiary education—through courses on data analytics, digital twins, simulation modeling, and intelligent transport systems—ensures that graduates are capable of designing and managing resilient, efficient, and sustainable waterborne logistics networks aligned with the objectives of smart mobility and green transition policies.

The project partners have expertise in the field of tertiary education related to inland navigation and logistics subjects. Based on their experience and interests for their students, the topic inland navigation and logistics has been divided between 4 partners, each partner contributing to the development of database containing subjects that will be updated or integrated in their curricula using AI technology. Their proposal for inland navigation and logistics topic can be found in Figure 3.

FH OÖ FORSCHUNGS & ENTWICKLUNGS GMBH	ZILINSKA UNIVERZITA V ZILINE
• Sustainable and Green Transport • Technical Equipmnet of Ports • Eco-Efficient Inland Waterway Logistics • Urban Green Logistics & Last Mile • Future Transport Ideas	• Maritime transport and Maritime Containers • Inland navigation and ports • Multimodal Transport • Waterways and Technical Equipment of Ports

Figure 3 Inland navigation and logistics

3.3 Communication Sciences

Artificial intelligence is becoming an important enabler in the field of Communication Science, as it reshapes the way messages are created, distributed, analysed, and adapted to different audiences. In both academic study and professional practice, AI can support a more strategic, data-informed, and interactive understanding of communication processes. One of the most relevant contributions is in the area of smart communication, where AI-based tools help organizations monitor audience behaviour, identify communication patterns, personalize content, and optimize the timing and channel of message delivery. Through predictive analytics, sentiment analysis, and automated content adaptation, communication can become more targeted, responsive, and efficient.

The subject of Communication Science will be significantly improved through AI-enhanced learning methods. For example, gamification supported by AI will make communication training more dynamic and engaging. Students will interact with simulations, adaptive quizzes, role-play scenarios, and intelligent feedback systems that adjust to their performance and learning pace. This will allow learners to practice persuasion, negotiation, public speaking, media literacy, and digital storytelling in more immersive ways. Using AI, the content of subjects will be focus on support crisis communication training by creating realistic

simulations of emergency situations, public scandals, misinformation campaigns, or institutional image risks, helping students learn how to react quickly, ethically, and strategically under pressure.

In addition, using AI in communication science will strengthen areas such as media analysis, social media communication, intercultural communication, branding, public relations, and strategic organizational communication. It can assist in detecting trends, combating disinformation, evaluating public reactions, and developing more evidence-based communication strategies. At the same time, teaching this subject with AI should also include critical reflection on ethics, transparency, manipulation risks, data privacy, and the responsible use of automated communication tools. In this way, AI does not replace human creativity and judgment, but enhances Communication Science by making it more interactive, analytical, practical, and relevant to the digital society.

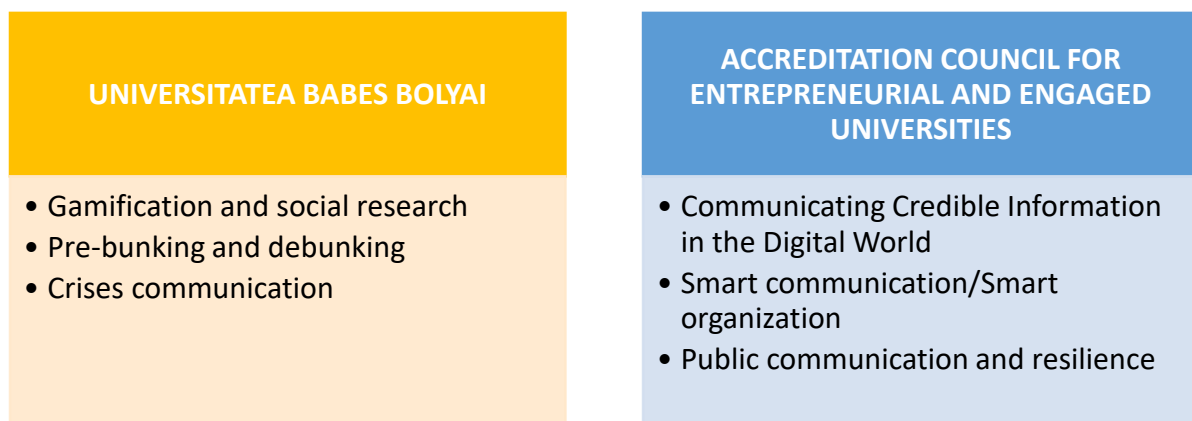


Figure 4 Communication Sciences

3.4 Health and Environment

Artificial intelligence offers major benefits for the development of academic subjects related to health and environment because it enables a more applied, data-driven, and interdisciplinary learning process. In health-oriented subjects, AI will support tertiary education through improvement of the understanding of clinical reasoning, diagnosis, treatment planning, and patient monitoring by exposing students to tools that increasingly shape modern healthcare practice. One relevant topic is AI and the Google Effect on clinical learning, where students rely less on memorizing large quantities of information and more on knowing how to search, verify, interpret, and apply digital knowledge responsibly. In this context, AI will help shift education from simple information recall toward critical thinking, evidence appraisal, and decision-making. Students will learn not only how to access medical information quickly, but also how to evaluate the reliability of AI-generated and online clinical guidance. Closely related subjects such as personalized medicine, digital health, telemedicine,

public health surveillance, and predictive prevention can also be significantly enriched through AI-based case studies and adaptive learning tools.

Using AI for the development of topics related to health and environment is also highly valuable in subjects dealing with human–AI interaction in clinical diagnosis and treatment. It will be used to simulate diagnostic pathways, compare probable diagnoses, interpret medical images, support treatment selection, and highlight risk factors based on patient data. This will help learners to understand both the strengths and limitations of AI in healthcare, while reinforcing the importance of human judgment, ethics, empathy, and accountability in clinical practice. In environmental and health-related interdisciplinary subjects, AI will support the study of medical waste management, resource optimization, and sustainability in healthcare systems. AI will be used to model waste generation, improve sorting and disposal processes, predict hazardous waste flows, and support greener hospital practices. It will also contribute to subjects such as environmental health, epidemiology, pollution monitoring, healthcare logistics, and climate-related health risks.

UNIVERSIDAD DE MALAGA
• AI and the "Google Effect" on Clinical Learning • Human-AI Interaction in Clinical Diagnosis and Treatment • Perceived Dehumanization in Digital Primary Care • Generative AI for the Creation of Clinical Simulation Scenarios • Detection of Sedentary Behavior Patterns using IoT and AI • Medical Waste

Figure 5 Health and Environment

4. Selection of disciplines/subjects

According to SMARTIE indicators, the project partners had to proposed at least 4 discipline/partner. A total of 36 disciplines/subjects have been foreseen to be proposed for updating process using AI tools. According to their expertise and interests, the project partners proposed a total of 38 subjects covering the SMARTIE topics as follows:

- Engineering: 17 subjects
- Inland navigation and logistics: 9 subjects
- Communication Science: 6 subjects
- Health and Environment: 6 subjects

The selection criteria of the 18 disciplines/subjects that will be updated in WP3 consisted of applying survey among SMARTIE partners. The analysis of the answers revealed the following subjects that have selected for improvement using artificial intelligence.

Table 1 Disciplines/Subjects selected for update using AI tools

Topic	Disciplines/Subjects	Responsible Partner
Engineering	Joining Processes of Advanced Materials	UCV
	Microwave Hybrid Heating of Materials	
	Weldability of the carbon steels	ULISBOA
	Adhesive bonding	
	Friction stir welding	UL
	Wire-arc additive manufacturing	
	Optimised parametric reconfiguration and automation systems	POLIBA
	Post processing Additive manufacturing in HRC	
Inland Navigation and Logistics	Sustainable and Green Transport	FHOO
	Eco-Efficient Inland Waterway Logistics	
	Maritime transport and Maritime Containers	UZ
	Waterways and Technical Equipment of Ports	
Communication Science	Public communication and resilience	ACEEU
	Communicating Credible Information in the Digital World	
	Gamification and social research	UBB
	Crisis communication	
Health and Environment	AI and the "Google Effect" on Clinical Learning	UM
	Medical Waste	

5. Conclusion

The analysis presented in this report confirms that artificial intelligence has significant potential to improve a wide range of university subjects across diverse disciplinary areas. The proposals developed by the SMARTIE consortium demonstrate that AI can support not only technical and engineering education, but also fields such as logistics, communication sciences, health, and environmental studies. In each of these domains, AI contributes to more data-driven, interactive, adaptive, and professionally relevant learning processes.

The report also highlights the strong engagement of the project partners in identifying subjects that are both academically relevant and strategically important for curriculum modernization. The fact that the number of proposed disciplines exceeded the initial target illustrates the high level of interest and the broad recognition of AI as a catalyst for educational innovation. Through the internal selection process, the consortium was able to establish a focused set of subjects that will be further updated in the next implementation phase, ensuring both thematic diversity and alignment with project objectives.